

Flatcar Container Linux Overview

Flatcar Container Linux is a minimal, immutable, container-optimized Linux OS. It ships only what is required to run container workloads, uses a read-only `/usr` partition, has no package manager, and updates itself automatically via atomic A/B partition swaps. It is the active successor to CoreOS Container Linux (EOL 2020) and a CNCF project.

Five Core Tenets

Tenet	Description
Immutable & image-based	<code>/usr</code> is read-only and dm-verity protected. No package manager. Updates replace all OS binaries atomically.
Minimal & container-optimized	Ships only Docker, containerd, and basic node tools. Applications run as containers or systemd-sysext images.
Fully automated	Declarative provisioning config applied once at first boot via Ignition. No configuration drift.
Tested & self-updating	100+ automated scenario tests. Atomic in-place updates with configurable reboot strategies. Atomic rollbacks.
Community stewarded	CNCF project. Maintainers from multiple organizations. Not vendor driven.

Update Mechanism

Flatcar uses the **USR-A / USR-B** dual-partition scheme (same as ChromeOS). One partition is active, one is standby. Updates are applied to the standby partition; a reboot switches to it. If boot fails, the system automatically falls back to the previous partition.

Update channels:

- **Alpha** — latest features, used by cutting-edge users
- **Beta** — release candidates, no known issues promoted from Alpha
- **Stable** — production-ready
- **LTS** — 18-month support window, new major release every 12 months

What Flatcar Includes

- Linux kernel + initrd
- Docker and containerd (as sysext images, opt-out)

- systemd, networkd, udev
- Ignition provisioning service
- Basic node tools: partitioning, crypto, volume management, networking
- update-engine (Omaha protocol update client)
- locksmithd (reboot coordinator; FLUO recommended for Kubernetes)

Nothing is installed at runtime — user applications run as container images or [systemd-sysext images][flatcar-provisioning].

Flatcar vs CoreOS Container Linux

Aspect	Flatcar	CoreOS CL
Status	Active	EOL (2020)
Drop-in replacement	Yes	—
Naming differences	flatcar.first_boot , flatcar.oem.id	coreos.first_boot , coreos.oem.id
Migration path	In-place update or fresh install	See flatcar-reference

Editions

Edition	Support	FIPS	Release Cadence	Security Updates
Flatcar (community)	Community	—	~2 months	Until next release
Flatcar Pro	Commercial (Kinvolk/Microsoft)	Soon	~2 months	Until next release
Flatcar LTS	Commercial	Yes	12 months	18 months

Documentation Sections

Section	Focus
flatcar-provisioning	Ignition, Butane/CLC, sysext, Terraform
flatcar-installing	Cloud providers, VMs (KubeVirt, QEMU, libvirt), bare metal
flatcar-setup	Clusters, customization, security, storage, systemd, updates, debugging

Section	Focus
flatcar-container-runtimes	Docker, containerd, Kubernetes, high-availability
flatcar-nebraska	Self-hosted update management server
flatcar-reference	Developer SDK, supply chain security, learning series, CoreOS migration

Community

- Slack: `#flatcar` in Kubernetes Slack (`kubernetes.slack.com/archives/C03GQ8B5XNJ`)
- Discord: <https://discord.gg/PMYjFUsJyq>
- Mastodon: `@flatcar@hachyderm.io`
- GitHub: <https://github.com/flatcar>